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Creating Sweet Spot in Tight and Unconventional Reservoirs Utilizing High Power Laser Technology

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Abstract

This paper presents the methods used to create sweet spots in tight formation and unconventional reservoirs utilizing proven methods of high-power laser technology. High power laser is already deployed successfully in the field. The technology is revolutionizing several upstream operations. The innovative technology was able to perforate a well create sweet spot without affecting the substrate integrity and damage.

The development of the first high power laser field system is based on several years of research and tool iteration, integrating optical, mechanical and sensing components in one tool that is configured to manipulate the beam at different orientation. The principle of the technology is to deliver the laser beam into the targeted formation, including tight formation, the controlled heat creates micro and macro cracks, dissociate, dehydrate and collapse minerals, resulting is enhancement in the permeability and creating sweet spot. Laser generator is located at the surface and the beam is transmitted via fiber topics cable to the tool downhole.

The technology can be applied in both open and cased hole, the first industrial high power laser perforation technology was successfully deployed in a cased well, laser technology can be applied in all rock types regardless their strength and mineralogy. In the cased hole, the downhole laser tool successfully penetrated through casing, cementing and formation, the candidate well is water disposal, and the objective is to improve the injectivity. The success of the operation was validated by field measurements including injectivity test, caliper and sonic logs, and downhole camera. The high power laser generator is located at the surface, and the beam is delivered via fiber optics cable to the target downhole.

The innovative technology was successfully deployed in different applications in the field with remarkable results, the technology is compact, versatile and sustainable it is a none explosive and waterless based technology, small footprint, and cost effective. The bean can be controlled very precisely to a specific target in any direction, as the technology is stress independent.

Sweet Spot in Unconventional Reservoirs: Identification & Challenges

In the case of unconventional reservoirs, a sweet spot can refer to a specific region within a reservoir that exhibits optimal conditions for hydraulic fracturing and hydrocarbon extraction. These areas are

identified with a variety of reservoir parameters, including lithological and petrophysical properties as well as geomechanical features, such as stress regimes and fracture orientations (Rahman & Gui, 2016; Glaser et al., 2014).

Reservoir quality parameters play a crucial role in sweet spot identification. These parameters include total organic content (TOC), which indicates the presence of hydrocarbons, as well as porosity and permeability. Additionally, the thickness of the pay zone, mineralogy, thermal maturity, and natural fracture density are considered. These parameters help in understanding the heterogeneity of the reservoir and identifying potential locations for development (Rahman & Gui, 2016; Glaser et al., 2014).

This paper will focus on a number of these parameters, namely; heterogeneity, reservoir completion quality, natural fracture network, and geomechanical properties, which can be affected/manipulated by high-power laser treatment.

One of the primary challenges is the heterogeneity of shale reservoirs. Organic shales are characterized by significant variability in their geological properties and hydrocarbon content. This heterogeneity makes it difficult to identify consistent sweet spots across the reservoir. Parameters such as total organic content (TOC), porosity, and permeability can vary widely, adding complexity to the characterization and development of these reservoirs, as well as perforation these areas with conventional methods. Another major challenge is reservoir completion quality. Achieving high-quality reservoir completions in organic shales is challenging due to the low permeability and brittleness of the rock. Traditional completion techniques may not be effective, and new technologies and methods are needed to optimize hydraulic fracturing (Glaser et al., 2014).

The presence and behavior of natural fracture networks also pose a significant challenge. Natural fractures can significantly influence the productivity of organic shales, they can be small-scale and hard to detect with conventional tools. The existence of a network of natural fractures directly impacts hydraulic fracturing, hence, is vital for optimizing production. Additionally, the geomechanical properties of organic shales play a critical role in determining the effectiveness of hydraulic fracturing. Properties such as Young's modulus and Poisson's ratio can vary significantly within the reservoir, making it challenging to design optimal fracturing treatments, especially, conventional ones.

High Power Laser Applications

High-power laser (HPL) presents a multitude of applications in upstream ranging from surface functions to drilling & completions. One of which is HPL perforation, a pioneering approach that offers a viable alternative to conventional methods by inducing rapid and localized thermo-mechanical processes, thereby altering the physical and chemical structure of rocks and their associated properties. This technique employs a multikilowatt, continuous wave (CW) laser, typically operating within the near-infrared spectrum, to induce chemical and physical transformations such as spallation, dissociation, melting, or vaporization of the rock, as well as catalysis of chemical reactions (Batarseh et al., 2023; Batarseh et al., 2019a; Batarseh et al., 2017).

A significant advantage of high-power laser perforation lies in its ability to operate independently of the stresses within the formation, in contrast to mechanical processes. This independence enables precise control over the direction of the laser, and consequently, the perforation. As a result, users can target any wellbore region and create a tunnel in any orientation, facilitating the design of complex perforations (Batarseh et al., 2019a; Batarseh et al., 2017).

High Power Laser & Sweet Spot Optimization

The fundamental concept involves utilizing HPL to generate micro-cracks and induce pyrolytic alterations in rock formations, thereby reducing the breakdown pressure. This approach can decrease the need for water and chemicals in hydraulic fracturing processes and mitigate the risks of uncontrolled fracturing and

associated safety hazards. The intensity and nature of the impact is influenced by various environmental and laser parameters, such as the laser's energy distribution, power, exposure duration, and the geomechanical state of the formation. For instance, the laser can induce thermal stress, resulting in thermal cracking and spalling. It can also initiate thermal decomposition (pyrolysis) and phase changes, altering the rock's composition and geomechanical properties within and around the heat-affected zone (HAZ).

HPL Effect Per Rock Type & Sedimentary Features

The thermal characteristics of laser perforation make it independent of rock type, stress, hardness, fractures, lamination, or any heterogeneity. The laser beam penetrates straight and symmetrically, unaffected by geomechanical stresses or material variations. For example, CT images (Figure 1) before and after laser perforation on both yellow limestone and highly laminated shale demonstrate a uniform and symmetrical tunnel. The laser beam passes through laminations without deviating, showcasing its precision and consistency (Batarseh et al., 2019a). This uniformity is crucial for creating consistent sweet spots and enhancing reservoir connectivity, regardless of the rock's inherent properties.

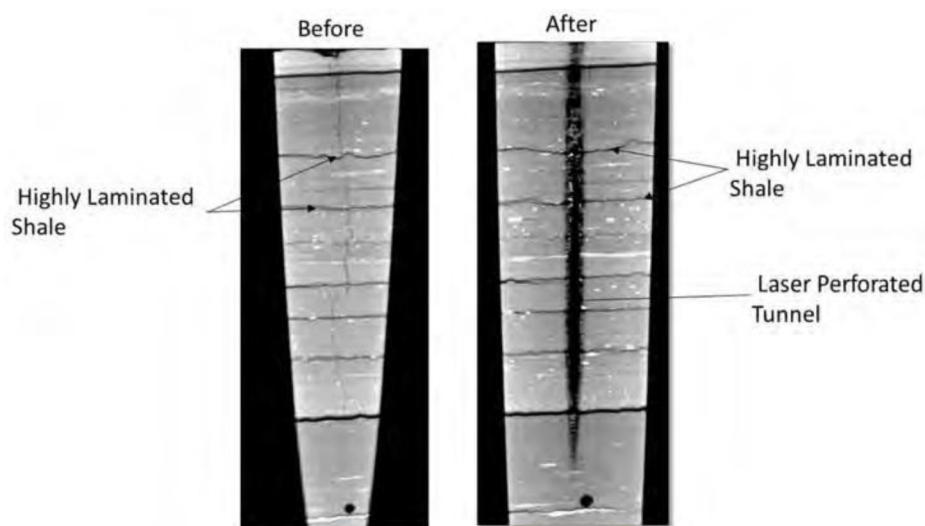


Figure 1—Uniform laser perforation tunnel shape in highly laminated shale (Batarseh et al., 2019a).

Although incurred effect might be the same, however, delivery parameters could defer from one rock type to another as absorption and diffraction processes also differ depending on surface and environment characteristics of the laser target. The absorption and reflectance of high-power laser energy by rocks are influenced by various factors, including heterogeneities, polycrystalline structure, morphology, and impurities such as organic matter and water content (Batarseh et al., 2019b). As a result, even similar rock types can exhibit varying absorption and reflectance properties. Previous studies have shown a correlation between the average absorption of rocks, measured using the Kubelka-Munk factor, and their total organic content (TOC) (Herron et al. 2014).

For this application, understanding the absorption properties of unconventional rocks is crucial, as it determines how the laser energy is absorbed and propagates through the rock. For instance, certain rocks like anhydrite and some shales exhibit high absorption, while others, such as yellow limestone and some shales, display low absorption (Figure 2) (Batarseh et al., 2019b). During perforation, the laser beam encounters multiple layers of casing, comprising steel and cement layers of varying thickness, as well as rocks with different saturation levels and conditions. Consequently, the beam's absorption and penetration rate vary at each stage. Additionally, environmental conditions can impact the interaction between the laser and the rock,

further influencing the outcome. Even within the same sample of shale, varying TOC produces different reflectance patterns as in Figure 3 (Batarseh et al., 2019b).

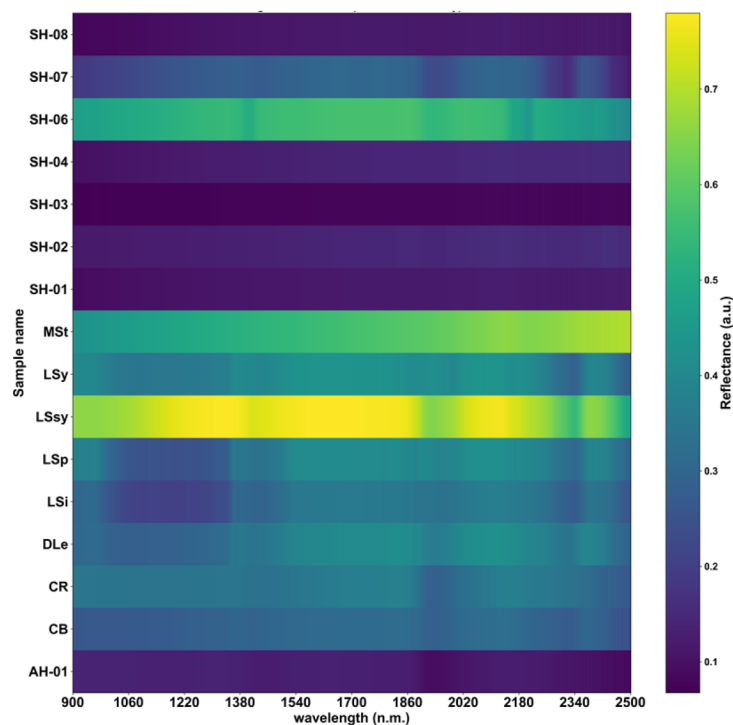


Figure 2—Measured reflectance for different rock types: anhydrite (AH), calcium carbonate (CB), concrete/cement (CR), dolomite limestone (DLe), Indiana limestone (LSi), desert pink limestone (LSp), yellow limestone (LSy), Saudi yellow limestone (LSsy), steel (MSt), and shales (SH) of varying origins (Batarseh et al., 2019b).

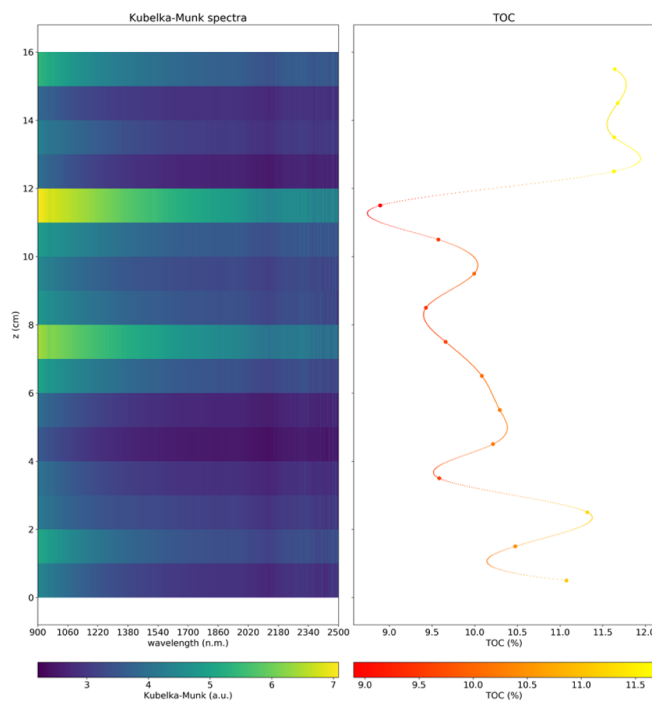


Figure 3—Kubelka-Munk spectra for a shale sample exhibiting varying levels of total organic content (TOC) (Batarseh et al., 2019b).

Permeability Enhancement

Al-Badairy et al. (2019) examined the effect of utilizing high-power laser on a formation's permeability. A laser beam with a 1.5 kW power output was used to scan the center of a sandstone sample (see Figure 4). For comparison, the sample's permeability is assessed across the pre- and post-treated areas of the sample. The permeability AutoScan depicts the behavior of permeability throughout the sample following laser treatment; permeability is increasing as the measurement gets closer to the center of the sample, where the sample was exposed directly to the high power laser beam (see Figure 4). The treated area's permeability was found to have significantly increased. Batarseh, Harith & Othman (2019) attributed this impact to the abrupt rise in temperature, which results in a thermal shock and micro-fractures in the sample. The study also suggested that clays will collapse at higher temperatures (i.e., 700 °C), increasing the void area, and if these are interconnected, then all sorts of rocks, including unconventional rocks, will have improved permeability near the perforation tunnels.

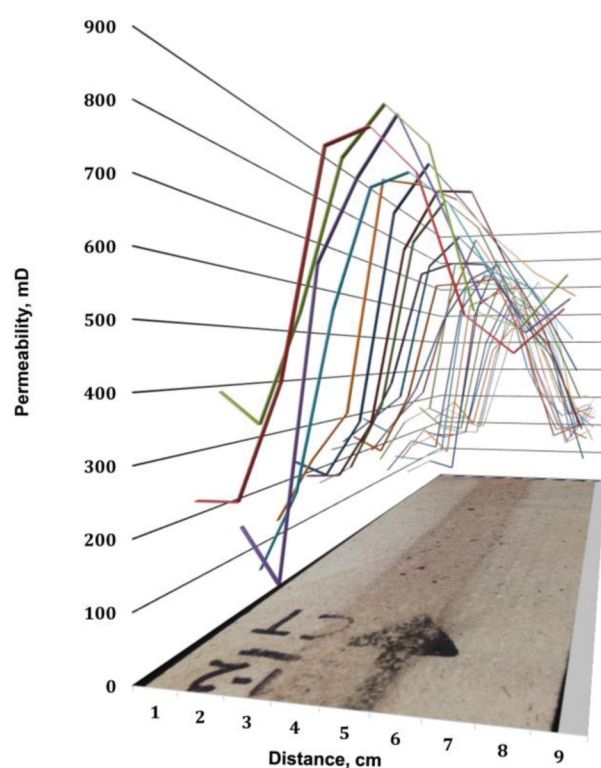


Figure 4—Effect of utilizing high-power laser on a formation's permeability (Al-Badairy et al., 2019)

In the same study, an unconventional rock sample was also tested. The experiment utilized three laser power levels; 1 kW, 2 kW, and 3 kW, to assess how varying power affects the rock and to determine the specific energy required for formation stimulation. Permeability was measured at 20 points along a straight line in the middle of the sample before and after treatment. Figures 5a and 5b illustrate the sample's condition prior to and following the laser application. Results show positive correlation between laser power and permeability, with higher power leading to increased permeability with the 3 kW treatment demonstrating the highest permeability improvement post-treatment. This enhancement is attributed to the abrupt temperature rise caused by the laser energy, inducing thermal shock and creating micro-fractures in the sample (Al-Badairy et al., 2019). Thermal imaging showed surface temperatures exceeding 1000°C, as depicted in Figure 6. These micro-fractures provide new pathways for fluid flow, thereby increasing permeability.

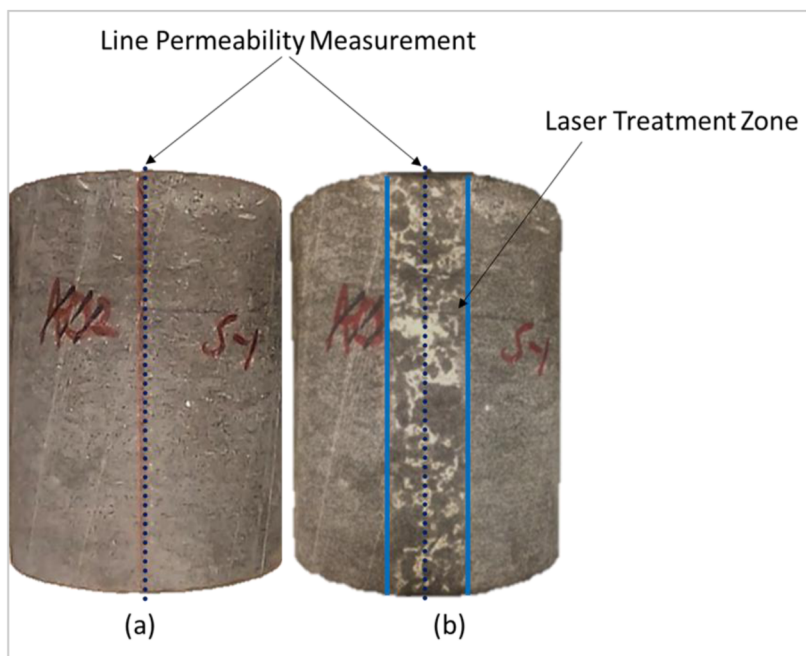


Figure 5—(a) Unconventional core sample before laser treatment, (b) The core sample after the treatment (Al-Badairy et al. 2019).

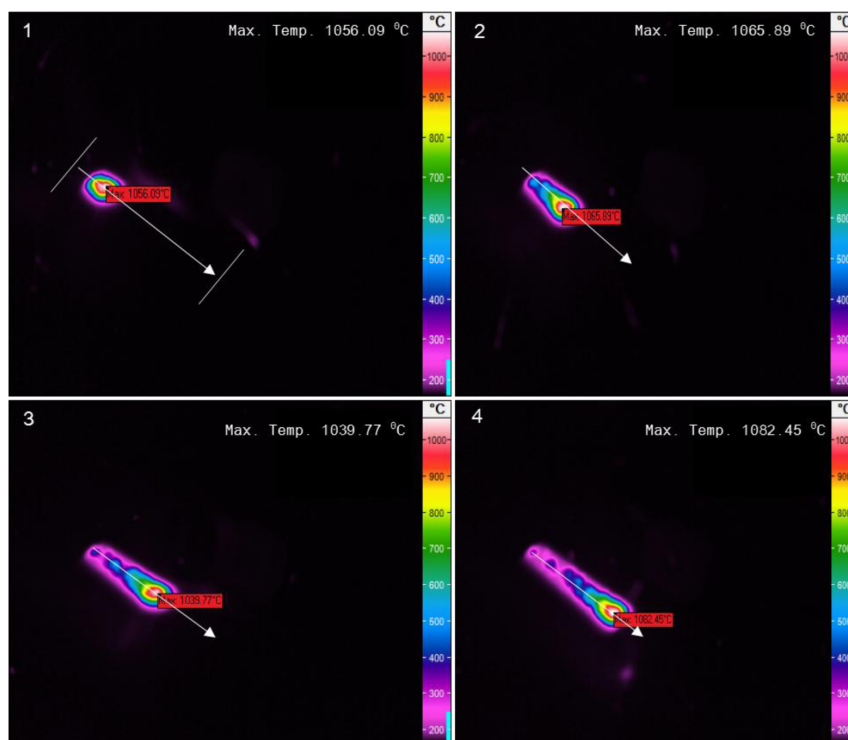


Figure 6—Thermal imaging of an unconventional core sample in different time-stamps during laser treatment (Al-Badairy et al., 2019).

Al-Badairy et al. (2019) conducted further analysis on a shale sample, where permeability was measured using a grid set-up. This approach aimed to evaluate the laser's effect on the entire sample and its surrounding areas that are not directly exposed to the beam. The results, presented in Figure 7, show pre-treatment (A) and post-treatment (B) permeability maps, indicating improved permeability in darker red areas. The laser's impact extended from the center to the edge of the core.

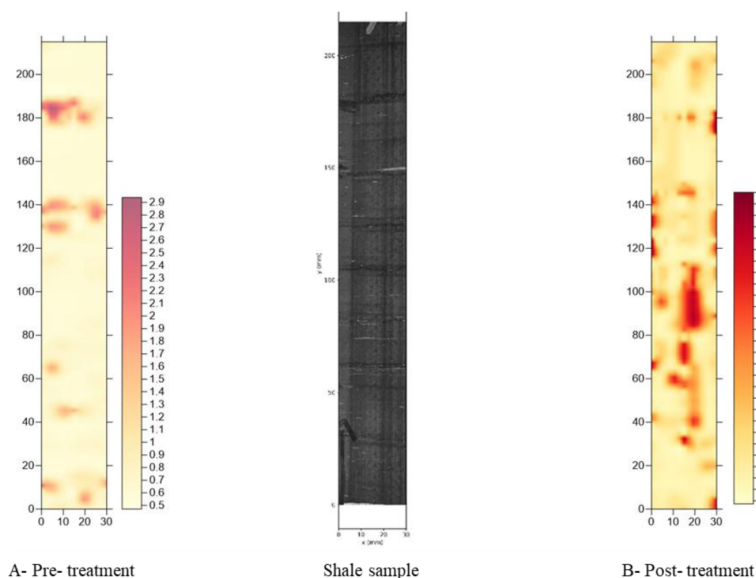


Figure 7—Permeability mapping of a shale sample pre- and post-treatment (Al-Badairy et al. 2019).

Induced Micro-Fractures

Other experimental studies have also shown that laser perforation can induce fractures in surrounding rock environment due to thermal stresses generated at the point of laser-rock interaction. These stresses enhance rock permeability and assist in initiating fractures. A thermally-induced fracture occurs when the thermal stress exceeds the rock's ability to withstand it (Miao et al., 2023), and this behavior is largely influenced by factors such as mineral composition, thermal properties, grain voids, and the magnitude of applied stress. When stress is applied to rock samples, the formation of macro-fractures becomes more likely (Batarseh et al., 2019a). As illustrated in Figure 8, a limestone sample exhibited fracturing under stress following laser perforation. Both macro- and micro-fractures resulting from exposure to laser may contribute to more favorable conditions for hydraulic fracturing.

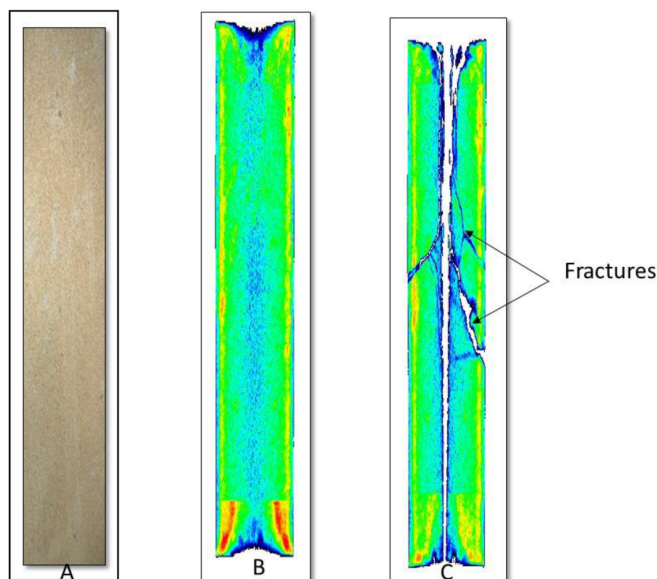


Figure 8—CT scan of laser-induced fracture in a limestone sample after HPL perforation (Batarseh et al., 2019a).

Thermal effects

Induced fractures and permeability increase are both a result of the thermal effects induced by laser. These thermal effects trigger mineralogical transformations that enhance permeability and fracture networks. When laser energy interacts with rock, it generates intense localized heat, producing thermal stresses that exceed the mechanical limits of the formation, resulting in micro- and macro-fracturing. This thermal-fracturing mechanism is particularly effective in rocks with high clay or carbonate content, as clays collapse and carbonates dissociate under elevated temperatures, weakening the formation and promoting natural fracture development (Al-Badairy et al., 2019). These changes are mapped using infrared (IR) imaging, which captures heat propagation during laser-rock interaction and provides valuable insight into the spatial distribution of thermal zones, as shown in Figure 9 (Batarseh et al., 2020). Enhanced zones (sweet spots) created via laser-induced thermal stress offer improved reservoir contact and fracture initiation, making them highly favorable for hydraulic stimulation.

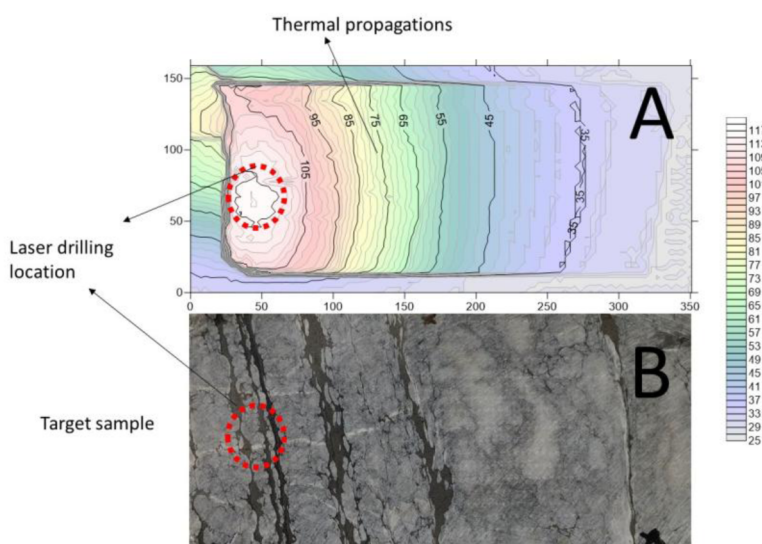


Figure 9—Infrared (IR) mapping illustrating: (A) the target sample surface, and (B) thermal propagation contours observed during the initial stage of laser treatment (Batarseh et al., 2020).

Thermal analysis techniques such as Thermogravimetric Analysis (TGA) and Differential Thermal Analysis (DTA) were also used to identify the temperature thresholds at which mineral transformations occur. For instance, TGA/DTA analysis on shale and limestone samples shows that clay minerals undergo dehydration and structural collapse between 450°C and 550°C, while calcium carbonate dissociates between 700°C and 900°C, depending on purity and composition (Al-Badairy et al., 2019; Somerton, 1992). Figure 10 demonstrates these behaviors clearly. This thermal breakdown not only weakens the rock but also increases porosity and permeability, key parameters for reservoir stimulation.

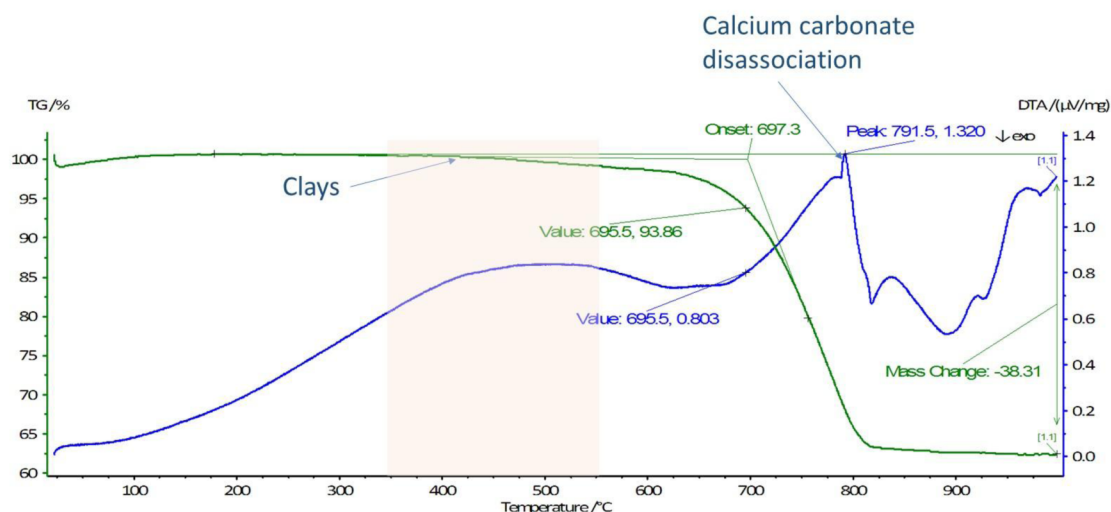


Figure 10—Thermal response of limestone at varying temperatures, highlighting the presence and breakdown of clay minerals, as well as the thermal decomposition of calcium carbonate (Al-Badairy et al., 2019).

Remarks & Way Forward

The application of high-power laser technology in unconventional reservoirs has demonstrated significant potential in improving rock flow properties, inducing fractures, and enhancing communication between the wellbore and the formation. By utilizing a high-power laser, it is possible to create a controlled hole with precise diameter, shape, and depth, regardless of the rock strength and maximum horizontal stress. The thermal energy generated by the laser weakens the formation, reducing its resistance to deformation and allowing fractures to initiate and propagate.

The ability to control the rock phase by adjusting the laser power enables the manipulation of the formation's properties, including vaporization, melting, and spallation. This level of control can be particularly beneficial in unconventional reservoirs, where the rocks often exhibit complex and heterogeneous properties. The use of high-power laser technology can reduce the breakdown pressure, leading to decreased water and chemical usage, as well as reduced energy consumption.

The creation of a near-wellbore sweet spot using high-power laser technology can be a promising approach for optimizing hydraulic fracturing in tight and unconventional reservoirs. By manipulating parameters such as heterogeneity, reservoir completion quality, natural fracture network, and geomechanical properties, it is possible to create a near-wellbore sweet spot that enhances reservoir connectivity and optimizes hydraulic fracturing.

The development and application of high-power laser technology in unconventional reservoirs require a comprehensive understanding of the coupled thermo-mechanical-electromagnetic interactions between the rock, fluids, and the laser. Further research and development should focus on optimizing the laser parameters, such as power and time, to achieve the desired rock phase and properties.

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